

Siluna Dangalla

Software Engineer · Systems & Security · Backend & App Development

✉ sdangalla44@gmail.com

☎ +94 740 478 458

🌐 github.com/GitGuru29

🌐 linkedin.com/in/siluna-dangalla-0744a02b1

🌐 silunadangalla.vercel.app

Colombo, Sri Lanka



ABOUT ME

I'm a final-year Software Engineering student with hands-on depth across systems programming, network security, backend development, and mobile and desktop application engineering. I don't just study software — I build it at a serious scale. My flagship project, NetSpector, is a solo-built network scanning and attack detection engine that grew over three years into a codebase on the level of Nmap or Wireshark. I designed AeroLang, a programming language for Android, from scratch — including the full compiler pipeline. I hold a Diploma in Cybersecurity and Ethical Hacking and apply that knowledge directly: from packet-level intrusion detection to OS-level privacy enforcement on macOS. On the AI side, I run CoreAI, a self-hosted inference system built on Ollama serving large language models locally over the network. I'm comfortable at every layer of the stack — from raw sockets and kernel extensions to REST APIs, mobile apps, and cloud deployment. I'm looking for an internship or entry-level role where I can contribute real engineering work from day one.

EDUCATION

BSc (Hons) Software Engineering

University · Final Year

Diploma in Cybersecurity and Ethical Hacking

Professional Certification

PROJECTS

NetSpector

C++ · Network Security

- Solo-built production-scale network scanning and attack detection engine rivalling Nmap/Wireshark in scope, developed over 3+ years
- Engineered deep-packet inspection, signature-based attack detection, and real-time traffic analysis across a large codebase
- Designed from scratch: raw socket layer, protocol parsers, detection rule engine, and reporting pipeline

AetherNet

C++20, Python, ONNX, Svelte · Hackathon

- High-performance user-space network virtualisation layer with ML-based adaptive proxy routing; competing in ProxyMaze '26 by Torch Labs
- Leading aether-core: TUN/TAP networking, Torch Proxies integration, and the full ML subsystem
- Trained traffic classifier and node congestion/failure predictor in Python; exported to ONNX and embedded into the C++20 runtime

CoreAI

Arch Linux · Ollama · On-Device AI Infrastructure

- Self-hosted AI inference system running large-parameter models (qwen2.5-coder:14b, deepseek-r1:14b) via Ollama on a dedicated local server
- Served over LAN for low-latency, privacy-preserving on-premise inference accessible from remote clients across the network

AeroLang

Compiler Engineering · aero-lang-web.vercel.app

- Designed and built an experimental Android-native programming language (v0.1.0-alpha) with Kotlin-inspired syntax

- Built the complete toolchain from scratch: lexer, parser, AST, type checker, and C++ codegen, plus formal grammar spec and test suite
- Shipped live website, interactive playground IDE, documentation, and public roadmap

PrivacyShieldMac

Swift · macOS System APIs

- macOS system-level privacy application that intercepts and controls application-level data exfiltration vectors
- Operates at the system extension boundary to enforce per-application privacy policies at the OS level

TECHNICAL SKILLS

Languages	C++, Python, Java, Swift, Kotlin, JavaScript, TypeScript, Bash, QML
Web / Backend	REST API design, Node.js, server-side architecture, backend systems
App Dev	Android (Kotlin/Java), iOS/macOS (Swift), cross-platform mobile & desktop
Systems	TUN/TAP networking, raw sockets, kernel/system extensions, user-space stacks
Security	Network scanning, attack detection, DPI, packet analysis, ethical hacking
ML / AI	PyTorch, ONNX, traffic classification, anomaly detection, on-device inference
Platforms	Linux (Arch, Ubuntu), macOS, Windows, Android
Tools	Git, CMake, Firebase, AWS, Cloudinary, Svelte, Vercel
Concepts	Compiler design, protocol engineering, proxy architectures, system design